Measuring Planck's and Boltzmann's constants through photoeffect and transistor collector-current

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The photoelectric effect was used to measure the Planck's constant as $h = (6.19 \pm 0.89) \cdot 10^{-34}$Js by analyzing the slope of the counter-voltages, where the photoelectric effect starts to occur, in respect to the frequency. Additionally, the Boltzmann's constant was determined to $k_B = (1.3913 \pm 0.0095) \cdot 10^{-23}$ Js by measuring the anti-diffusion current of a transistor at different temperatures in equilibrium.

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1 Introduction

In modern physics the SI-unit-system is based on seven fundamental constants. Two examples of these are discussed in this paper: One fundamentally important constant, which links energy and temperature scales, is known as the Boltzmann constant. In a sense, the Boltzmann constant is part of the central foundation of statistical physics as a proportionality factor between the micro-states of the closed system and the macroscopic quantity, the so called entropy. The Boltzmann constant plays another important role in the theory of ideal gases. With the law of ideal gases, the Boltzmann constant is a proportionality factor, which makes it possible to calculate the average thermal energy of atomic free particles with their temperature. Another fundamental constant in physics is the Planck constant, which is the ratio between the energy and the frequency of a photon. The discovery of this important quantity led to the foundation of today's quantum physics. Those mentioned constants will be examined in detail and also determined experimentally in the following sections.

2 Measuring Planck's constant

2.1 Theory: photoelectric effect

In order to understand the procedure for determining Planck's constant, it is important to understand the theory behind it. The basis of the following measurements is the so-called photoelectric effect. This describes that in general, sufficiently short-wave light dissolves negative charge out of carriers, i.e. electrons, from a metal surface. The higher the frequency of the dissolving light, the higher the kinetic energy of the dissolved electrons. The incoming photon energy can be described by the equation $E_{ph} = h\nu$. With this energy, the light particles hit the electrons located in the atom, which can absorb the incoming energy. If this energy is high enough, the electrons have a high enough energy W_A to overcome the atomic bond. The electrons are released from the metal and have a residual energy of:

$$E_{ph} = h\nu - W_A \tag{1}$$

This energy can be determined using a photocell. Photons release electrons from the cathode, which migrate to the anode. If one adds now an internal electric field opposite to the natural one, only those electrons arrive at the anode, which have a high enough kinetic energy to overcome the counter-voltage of the field. These arriving electrons are measured as photocurrent then.

2.2 Analysis of the Planck data

The experimental setup used is shown in figure 1. The basic setup consists of a photocell, a Hg lamp as photon source, different interference filters to obtain monochromatic light with different wavelengths. Hg has certain spectral lines with different wavelengths. To focus this incoming light, a lens was also placed directly in the focal point in front of the entrance to the photocell. The counter voltage was regulated with the help of the device ZENTRO electric 30V/1A and

the incoming current was detected with the highly sensitive Keithley 6458.

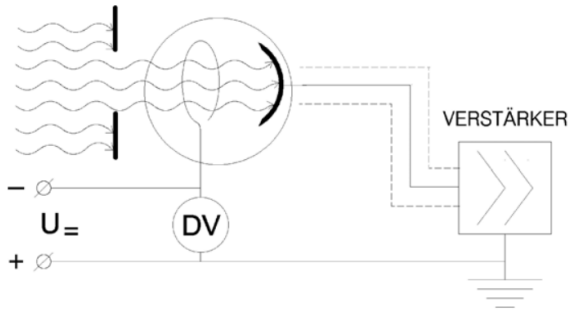

Fig. 1: Experimental setup to measure Planck's con- stant. Light from a Hg-lamp is focused with a lens onto the cathode of a photo cell, which is passing an interference filter on the way. The current at the cathode is then measured for different countervoltages between cathode and anode.

In all five curve plots, it is visible that for a while the characteristic curve runs in the negative current range. This can be explained by the fact that dirt very easily builds up on the anode and these dirt particles cause negative charge to generate a counter-current. This counter current is accelerated by the additional voltage applied and a negative electron current is recorded at high counter voltages. This opposing electron current causes a shift in the negative current direction for the entire curve. If the reverse voltage is now lowered, two things happen: On one hand, electrons dissolved by photons getting to the anode and on the other hand, the negative current dissolving from the dirt on the anode decreases, since the counter-voltage is now lowered originally acting as acceleration.

At a certain reverse voltage, called buckling-voltage, the curve increases strongly. At this voltage it is only possible for the electrons, which are heading directly to the anode, to overcome the reverse voltage. With this consideration it is possible to calculate the kinetic energy of the photo-electrons. The buckling-voltages of the five different wavelengths are listed in tab. 1. These values are now plotted against each other in Figure 3 in order to subsequently calculate the Planck constant, using the Einstein equation [1]. This is done by applying a regression line to the measured values. With the help of the slope m of this straight line, the elementary charge $e = 1.602176634 \cdot 10^{-19}$C 1can now be calculated by the equation $h = m\,e$. With this equation, the result of the experimental Planck constant was calculated to $h = (6.19 \pm 0.89) \cdot 10^{-34}$Js. This value agrees to 6.6% with the actual Planck constant of $h_{theo} = 6.62607015 \cdot 10^{-34}$Js 1.

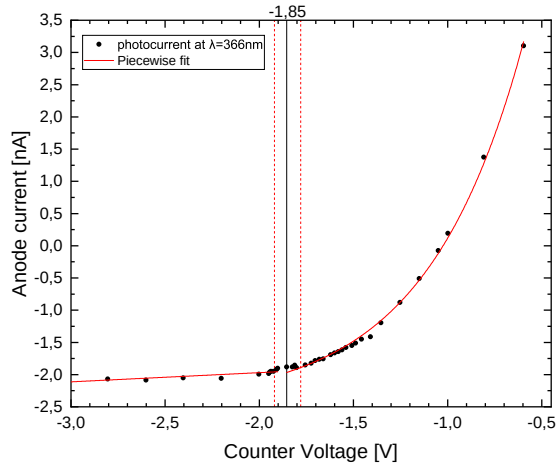

Fig. 2: Photo-current in dependence on the counter-voltage for $\lambda = 366$nm. A piecewise function is plottet through the data. First a linear plot, then there is used a exponential plot, after a specific counter-voltage.

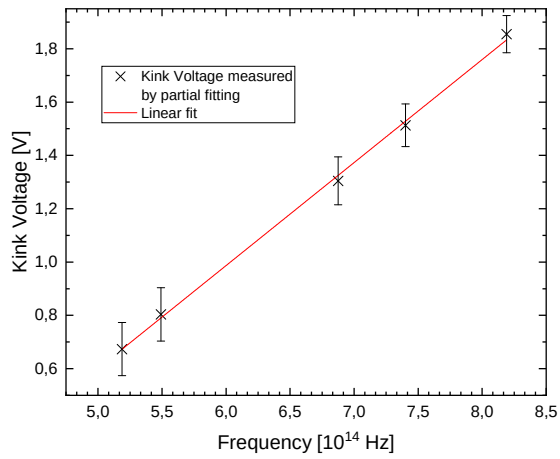

Fig. 3: The used counter-voltages of five different wavelengths plotted in one figure. Through the slope, which is $\frac{h}{e}$, we get the value of the experimentally determined Planck constant.

frequency $\nu/$ $[10^{14} Hz]$	counter voltage $U/$ [V]
5.187	0.7±0.1
5.491	0.8±0.1
6.876	1.30±0.09
7.402	1.51±0.08
8.191	1.85±0.07

Tab. 1: The frequencies of light, dissolving electrons from the kathode, and the specific corresponding counter-voltages are listed. At these voltages, it is possible for the first time that the fastest photo-electrons overcome the counter field and a photo-current is meassured.

Finally, the practically determined value agrees with the theoretically expected value within the error interval.

3 Measuring Boltzmann's constant

3.1 Shockley equation

In order to determine the Boltzmann constant experimentally, a semiconductor pn junction can be used, which is found in transistors and diodes. In such a transistor, even if no voltage is connected, an exchange of electrons takes place due to thermodynamic effects. This means that currents can flow even without voltage. Because these currents are in thermal equilibrium, the connection to the Boltzmann constant can be recognized and calculated.
The occurring current at such a junction can be described by the Shockley equation. In summary, the Shockley equation describes the current-voltage characteristic of a semiconductor diode. If you look at a semiconductor diode more closely, there are two areas. One is the p-contaminated side and the other is the n-contaminated side. At the junction of these two sides, a carrier-free zone is created, since negatively charged particles (free electrons) of the n-contaminated crystal and the positive particles (defect electrons) of the p-contaminated crystal diffuse to the other side and recombinate there. Further recombination is prevented by the stationary contaminated atoms forming a space charge as ions whose electric field counteracts recombination. The voltage of this built up field is called anti-diffusion-voltage. Due to temperature fluctuations, it is possible that bound electrons in the charge-free zone are lifted from the valence band into the conduction band and an internal current I_0 flows. In addition, there are also diffusing charge carriers that can overcome the anti-diffusion-voltage. This current is called diffusion-current I_D. In order to overcome the previous mentioned voltage, these particles must have the energy $E \geqslant e\,U_A$. Where e is equal to the elementary charge $e = 1.602176634 \cdot 10^{-19}\,\mathrm{C}$ 1. Since the energy of the electrons in the conduction band follow Boltzmann statistics, the diffusion current is proportional to the Boltzmann constant, with the formula

$$I_{D,0} \sim e^{-\frac{e\,U_A}{k_B\,T}} \tag{2}$$

If everything is now in thermal equilibrium, one can additionally assume the equation $|I_{D,0}| = |I_0|$. If an external voltage is now applied, the anti-diffusion voltage increases, from which the formula

$$I_D(U) = I_0\, e^{\frac{e\,U}{k_b\,T}} \tag{3}$$

follows. To obtain the total current, the internal current I_0 must be subtracted. The resulting equation corresponds to the Schockley equation. In reality, however, many interference factors can considerably influence the measurement, these interferences can be triggered by different currents inside. In order to prevent these influences, another semiconductor is now added, so that a npn transistor is created. As the internal current I_0 only overcomes the base-emitter junction, just the diffusion-current can be detected at the conducter output. By this consideration it is not necessary to use the actual Schockley equation, but equation [3] to describe the dependence of the collector current on the applied base-emitter voltage.

3.2 Analysis of Boltzmann's constant

The external voltage was applied to the emitter-base junction by a "centro electric" meter through a voltage divider. The current was measured by the multimeter "Agilent 34405A". The transistor used was the "npn- power transistor 2N3055". Throughout the experimental part, the transistor was kept constant at the same specific desired temperature by the "Lauda Ecoline Startedition RE104" thermostat. The I(U) curves were measured for temperatures $T_1 = 40°\mathrm{C}$, $T_2 = 80°\mathrm{C}$ and $T_3 = 120°\mathrm{C}$. The obtained measured data were subsequently plotted with a fit of an exponential function in a semi logarithmic plot. From the associated residuals, it can be seen that a fairly significant systematic error occurs at low and large voltages, where the data points do not really follow the expected exponential curve. This is due to the existing resistance of the emitter, which reduces the base-emitter voltage. This resistance can be compensated by changing equation [3] to:

$$I_D(U) = I_0\, e^{\frac{e\,(U - R\,I_C(U))}{k_b\,T}} \tag{4}$$

After equation [4] has been solved for $U(I_C)$, a logarithmic fit can now be inserted in order to determine the value of R. This procedure is illustrated in 4.

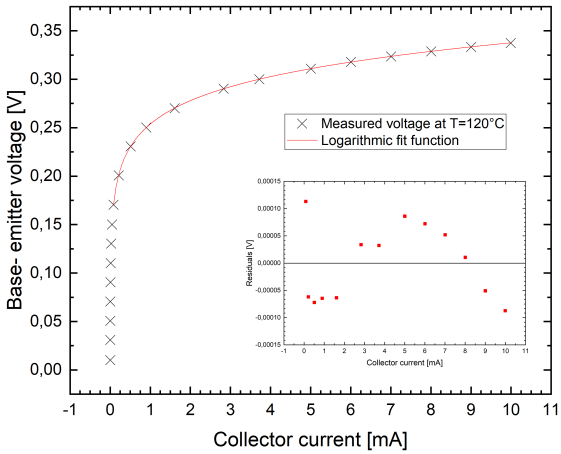

Fig. 4: Determination of the emitter-resistance by plotting the base-emitter voltage against the collector-current. The resistance is given by the parameters of a logarithmic fit function.

With this calculated value it is now possible to correct the values of the voltages U. These final values for the voltage and the current are now portrayed by the help of an exponential function in figure 5.

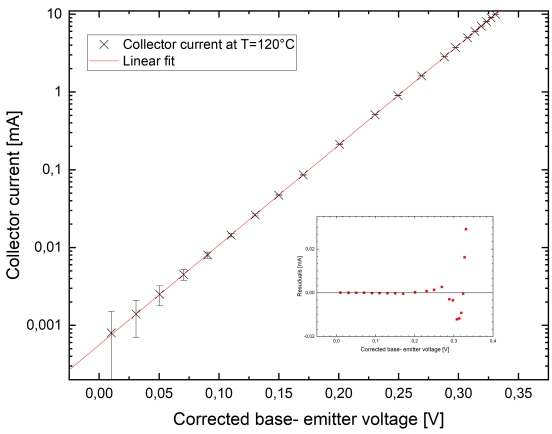

Fig. 5: Collector current in dependence on the corrected base-emitter voltage measured at temperature T = 120 °C in logarithmic application with a linear fit function. The slope of the linear fit can now be used to calculate the value of the Boltzmann's constant.

It is now possible to determine the Boltzmann's constant together with the associated error from the fit parameters. The three values of the Boltzmann constant are listed in 3.2. The experimentally determined mean value $k_B = (1.3913 \pm 0.0095) \cdot 10^{-23}$ Js deviates from the theoretically expected value $k_B = 1.380649 \cdot 10^{-23}$ Js by 0.76% . Still the experimentally determined value of the Boltzmann's constant does not coincide with the theoretical one. As the deviation is pretty small, we can acknowledge the Boltzmann's constant.

temp. T / [°C]	Boltzmann's const. $k_B/[10^{-24}\,Js]$
40	1.39135 ±0.00094
80	1.4049 ±0.0094
120	1.37758±0.00057

Tab. 2: The Boltzmann's constants, obtained at different temperatures, are listed above.

4 Results

In this paper, the fundamental constants of Planck and Boltzmann were experimentally determined. For the Planck constant, the photoelectric effect was used to determine the counter-field voltage at different light frequencies. After plotting the frequency towards the counter-voltage, the Planck constant was calculated to $h = (6.19 \pm 0.89) \cdot 10^{-34}$ Js using the slope, which is theoretically composed of $\frac{h}{e}$. The deviation to the theoretically expected value is 6.6%. However, due to the fact that the practically determined value here agrees with the expected one within the error, it can be said that this method is very well suited to determine this constant. Subsequently, the Boltzmann constant was determined. This value was determined by measuring the diffusion current inside a transistor at the pn junction. Due to the thermal equilibrium, the current follows the Boltzmann factor. Therefore, the data were interpolated and the constant was determined to $k_B = (1.3913 \pm 0.0095) \cdot 10^{-23}$ Js by the fit parameters.

References

[1] Meschede, Gerthsen Physik, 23, 2005

[2] Frauenfelder und Huber, Einführung in die Physik II,

[3] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 45

5 Appendix

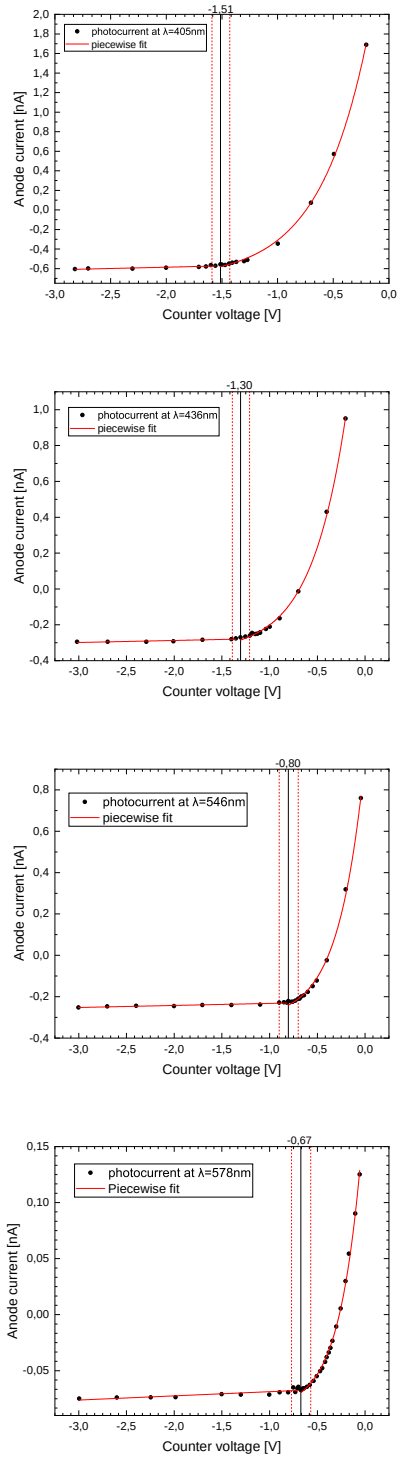

Fig. 6: Listed from above a)- d). Photocurrent in dependence of the countervoltage for light of the different wavelengths used. First linear, then exponential fits were used to determine the points where the relevant signal is detectable against the underground. Those points are the transition points between the two functions.